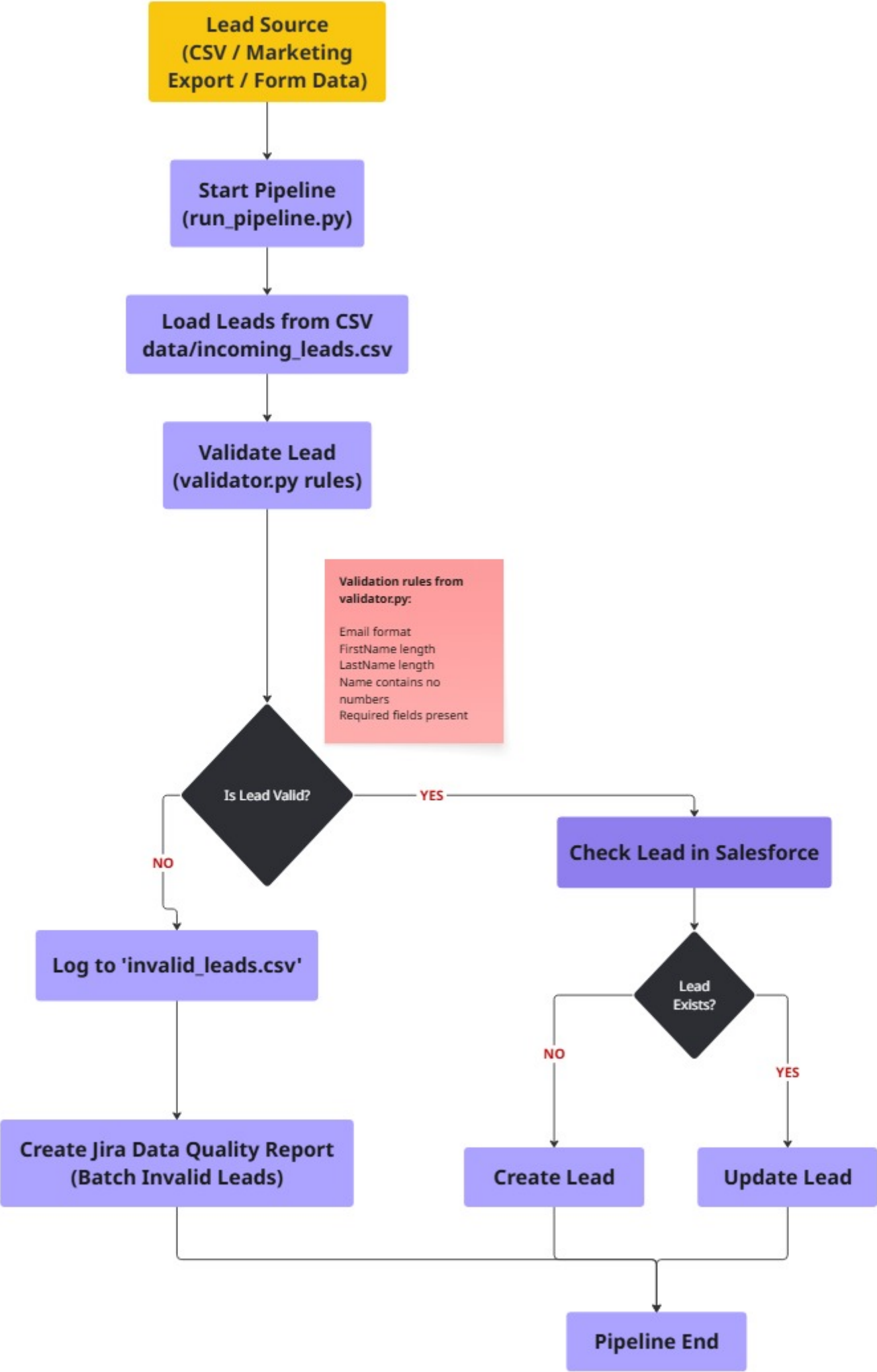


SOP: Salesforce Data Integrity & Validation Pipeline



SOP: Salesforce Data Integrity and Validation Pipeline

Purpose

The purpose of this pipeline is to ensure that all incoming lead data is validated before being inserted into Salesforce. Data quality issues such as incorrect email formats, missing fields, or invalid names can reduce the reliability of CRM data and affect marketing and sales operations. This pipeline automatically validates lead records, logs invalid entries, and reports issues through Jira to maintain Salesforce data integrity.

Data Sources

Lead data is received through CSV files that simulate common CRM data sources such as marketing lead exports, web form submissions, or sales lead imports. The pipeline reads lead records from the file located at `data/incoming_leads.csv`. Each record typically contains fields such as `FirstName`, `LastName`, `Company`, and `Email`.

Validation Rules

Before any record is processed in Salesforce, the pipeline validates the lead using a set of rules defined in the validator module. These rules ensure that required fields are present and correctly formatted. The validation checks include verifying that the email address follows a valid format, ensuring that first and last names meet minimum length requirements, confirming that names do not contain numeric characters, and verifying that required fields such as `Email`, `FirstName`, `LastName`, and `Company` are present.

Error Handling

If a lead fails validation, it is not inserted into Salesforce. Instead, the pipeline records the invalid entry in a file named `data/invalid_leads.csv` along with the validation error message. This file serves as an audit log of problematic records. After processing all leads in the batch, the pipeline generates a Jira ticket that summarizes the invalid records so that they can be reviewed and corrected.

Salesforce Processing

Valid leads are checked against Salesforce to determine whether the lead already exists. The pipeline performs a query using the Salesforce REST API to search for an existing lead with the same email address. If a matching lead is found, the existing record is updated with the latest data. If no matching lead exists, the pipeline creates a new lead record in Salesforce.

Automation Output

The pipeline produces several outputs during execution. Valid leads are created or updated in Salesforce. Invalid leads are stored in the file `data/invalid_leads.csv` for tracking and review. A Jira ticket is automatically generated containing a summary of the invalid leads detected during the batch run.

Technologies Used

The pipeline is implemented in Python and integrates with external systems using APIs. It uses the Salesforce REST API to create and update lead records and the Jira REST API to generate issue tickets for data validation failures. Lead data is processed using CSV file handling and validation logic implemented in Python.